AI-Augmented Marketing: a reusable skill library for marketing operations

Personal project, October 2026 | Chirag Arora | Python, Claude skills (SKILL.md), Markdown

Goal

Marketing teams repeat the same tasks every week: reading funnel data, writing updates, editing newsletters and training colleagues. I built four small, reusable Claude skills that cover these tasks, with scripts for the parts that must be exact, and tested them on a real public lead dataset. I built them with Claude as a coding assistant and checked the outputs myself.

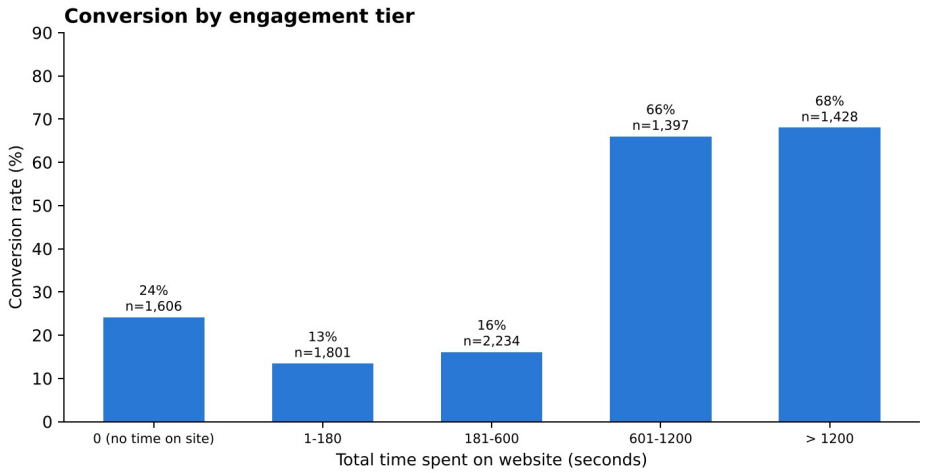
The four skills

Skill	What it does	How it was tested
funnel-kpi-report	Turns a lead CSV into conversion by channel and engagement tier, with 95% confidence intervals, plain-English insights and charts.	Run on 9,240 real leads, plus a digital-only run (8,466); unit test on synthetic data with known rates.
weekly-marketing-update	Renders a one-page update (headline, signal to act on, decisions, actions, risks) from the results and a CSV action tracker with overdue checks.	Generated from the real results; tracker overdue logic unit-tested.
newsletter-qa	Lints a draft: subject length, word count, reading grade, long sentences, one call to action, buzzwords, sourced numbers.	First draft failed 4 of 7 checks; revised draft passes 7 of 7. Lint unit-tested.
training-module-builder	Template and rules for a short hands-on module with goals, timed agenda, exercise, quiz and follow-up.	One 45-minute module written; not yet delivered to learners.

What the real data showed

Data: public X Education lead-scoring dataset, 9,240 leads. Manually added leads (offline referrals) converted at 92-99%, so I analysed the 8,466 digital inbound leads separately (overall conversion 34.1%).

- **Time on site is the strongest signal.** Leads with more than 600 seconds on site convert at 67% (n=2,825) vs 15% (n=4,035) for those with up to 600 seconds. A simple rule to decide who sales calls first.
- **Channels differ, but less.** Google 40%, organic search 38%, direct traffic 32%, chat 25% (95% intervals in the report). Chat follow-up needs work before more spend.
- **Email hygiene matters.** Leads whose email bounced converted at 5% (n=311); leads marked do-not-email at 12% (n=689).
- **Small groups are not interpreted.** Groups under 100 leads are merged, so the report does not over-read a high rate on a tiny sample.



Quality checks

- Four automated tests pass. They cover the confidence-interval maths against a known value, funnel rates and small-group merging on synthetic data with known answers, overdue-action logic and the newsletter lint.
- Every number in the sample newsletter and weekly update was checked against results.json.

Limits

- The data comes from an online-course provider (B2C) with no dates, so it shows associations, not causes, and no weekly trend. A B2B funnel with accounts and dates would need new column mappings and stage definitions.
- Owners in the sample tracker are placeholders. The training module has not been delivered to learners. The scripts are small prototypes, not production tools.
- The skills check form and arithmetic. A person must still verify claims and judge tone.

Next steps

- Connect to a CRM export with dates, add account-level (ABM) stages, and test the 600-second rule against live data before using it to route leads.

Source: public GitHub copy of the Kaggle Lead Scoring dataset (saptarshim7/Lead_Scoring_Case_Study, Leads.csv). Code, tests and outputs are in the project folder.